Product-Level Markdown Pricing Simulator

A Multi-Model Demand Estimation and MDP-Based Discount Optimization Framework

Applied to the Dunnhumby Complete Journey Dataset

Pricing Analytics Research

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Abstract

We develop a product-level markdown pricing simulator grounded in a Markov Decision Process (MDP) formulation where the state vector encodes per-item selection, discount depth, realized demand, and cumulative budget consumption, while the action space specifies the new selection and discount vector across all products. Using the Dunnhumby Complete Journey dataset (2.6M transactions, 92,339 products, 102 weeks), we estimate individual demand models for 128 representative products selected proportionally from 12 data-driven segments. We compare five demand model families—log-log OLS, Random Forest, XGBoost, LightGBM, and neural networks—finding that XGBoost achieves the lowest weekly aggregate MAPE (7.9%) while the classical log-log OLS attains the strongest log-space correlation ($r = 0.946$) and best discount-lift prediction ($r = 0.312$, 88% directional accuracy). Empirical Bayes shrinkage toward segment medians regularizes noisy per-product estimates. The calibrated simulator recovers monotone demand–price relationships for 11 of 12 segments and identifies a front-loaded discount policy that improves revenue by 72.6% over the no-discount baseline. Per-product sensitivity analysis reveals that 31.2% of products benefit from discounting, with a mean optimal discount depth of 13.9%.

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1 Introduction

Markdown pricing—the practice of reducing prices on products over their selling season—is one of the most consequential levers in retail revenue management. Retailers frequently rely on heuristic discount rules (e.g., uniform 20% off) that ignore heterogeneity in price elasticity across products. A principled approach requires three ingredients: (i) accurate demand models that capture how individual product sales respond to price changes, (ii) a simulation environment that faithfully represents the sequential nature of markdown decisions under budget constraints, and (iii) a policy evaluation framework that compares alternative discount strategies.

This work addresses all three requirements. We begin with a thorough exploratory analysis of the Dunnhumby Complete Journey dataset—a publicly available retail panel covering 2,500 households, 92,339 products, and 102 weeks of transactions. We then estimate individual demand functions for 128 representative products using five distinct model families, ranging from classical econometric specifications to modern machine learning methods. The estimated demand models feed a product-level MDP simulator whose state and action spaces operate at the individual item level, not at the category or segment level.

1.1 Related Work

Price Elasticity Estimation. The log-log (constant-elasticity) demand model has deep roots in empirical economics. Tellis [1] surveyed 367 elasticity estimates and found a mean price elasticity of -1.76. Bijmolt et al. [2] extended this meta-analysis and confirmed that constant-elasticity specifications remain competitive with more flexible functional forms for aggregate demand. Our baseline log-log OLS model directly implements this specification at the individual product level.

Machine Learning for Demand Estimation. Ferreira et al. [3] demonstrated that machine learning models (random forests, gradient boosting) can outperform classical econometric models for retail demand prediction, especially when the price–demand relationship is nonlinear. Ban and Rudin [4] applied gradient-boosted trees to the newsvendor problem, showing that ML methods can capture complex feature interactions that linear models miss. Chen and Guestrin [5] introduced XGBoost, which has become a standard tool for tabular prediction tasks. Ke et al. [6] proposed LightGBM with histogram-based splitting for faster training on large datasets.

Empirical Bayes Shrinkage. When estimating parameters for many products individually, some estimates will be noisy due to limited data. The James–Stein estimator and its extensions [7, 8] show that shrinking individual estimates toward a grand mean dominates the unbiased estimator in terms of total squared error. Morris [9] formalized the parametric empirical Bayes framework. We apply this principle by shrinking product-level elasticities and discount effects toward their segment medians, with shrinkage weights determined by each product model's R^2.

Revenue Management and MDP Formulation. Our MDP formulation follows the framework of Phillips [10] and Talluri and van Ryzin [11], who established the theoretical foundations for pricing and revenue management under sequential decision-making. The

simulator implements a finite-horizon discounted MDP with budget constraints, analogous to the markdown optimization models studied in retail operations research.

Deep Learning for Demand. Hartford et al. [12] proposed Deep IV for counterfactual demand estimation using instrumental variables within a neural network architecture. Gabel and Timoshenko [13] developed scalable deep learning models for product choice with large assortments. While these approaches show promise for very large datasets, our experiments show that small per-product sample sizes (50–80 observations) make neural network estimation unreliable without pooling across products.

1.2 Contributions

1. **Product-level MDP simulator** with exact per-item state and action vectors, operating at the SKU level rather than the category or segment level.
2. **Systematic comparison of five demand model families** (log-log OLS, Random Forest, XGBoost, LightGBM, neural network) with empirical Bayes shrinkage, evaluated on held-out test data.
3. **Comprehensive policy evaluation** comparing seven discount strategies including uniform, targeted, front-loaded, and ROI-optimized policies.
4. **Per-product sensitivity analysis** identifying which products benefit from discounting and their optimal discount depths.

1.3 Paper Organization

Section 2 describes the dataset and exploratory analysis. Section 3 presents the product segmentation and representative selection methodology. Section 4 details the five demand model specifications and the empirical Bayes shrinkage procedure. Section 5 formalizes the MDP simulator. Section 6 presents the model comparison, validation, policy evaluation, and sensitivity analysis. Section 8 concludes.

2 Data and Exploratory Analysis

2.1 Dataset Overview

The Dunnhumby Complete Journey dataset records household-level transactions from a major US retailer over a two-year period (104 weeks). The data comprises seven interrelated tables:

Table 1: Dataset composition.

Table	Description	Records
`transaction_data`	Item-level purchases	2,595,732
`product`	Product attributes (department, category)	92,339
`hh_demographic`	Household demographics	801
`causal_data`	Store–product–week promotions	68,377,428
`coupon`	Coupon–product associations	1,135
`coupon_redempt`	Coupon redemptions	2,318
`campaign_desc`	Campaign metadata	27
`campaign_table`	Household–campaign assignments	4,450

After restricting to the stable period (weeks 18–102) to avoid start-up transients, we reconstruct base prices and discount depths from the transaction records:

$$\text{base_price}_t = \text{sales_value}_t + |\text{retail_disc}_t| + |\text{coupon_disc}_t|, \tag{1}$$

$$\text{discount_depth}_t = \frac{|\text{retail_disc}_t| + |\text{coupon_disc}_t|}{\text{base_price}_t}. \tag{2}$$

2.2 Temporal Patterns

Figure 1 shows weekly aggregate revenue and transaction counts over the observation period. Revenue exhibits strong weekly seasonality with holiday peaks. The stable period (weeks 18–102) provides 85 weeks of consistent purchasing behavior.

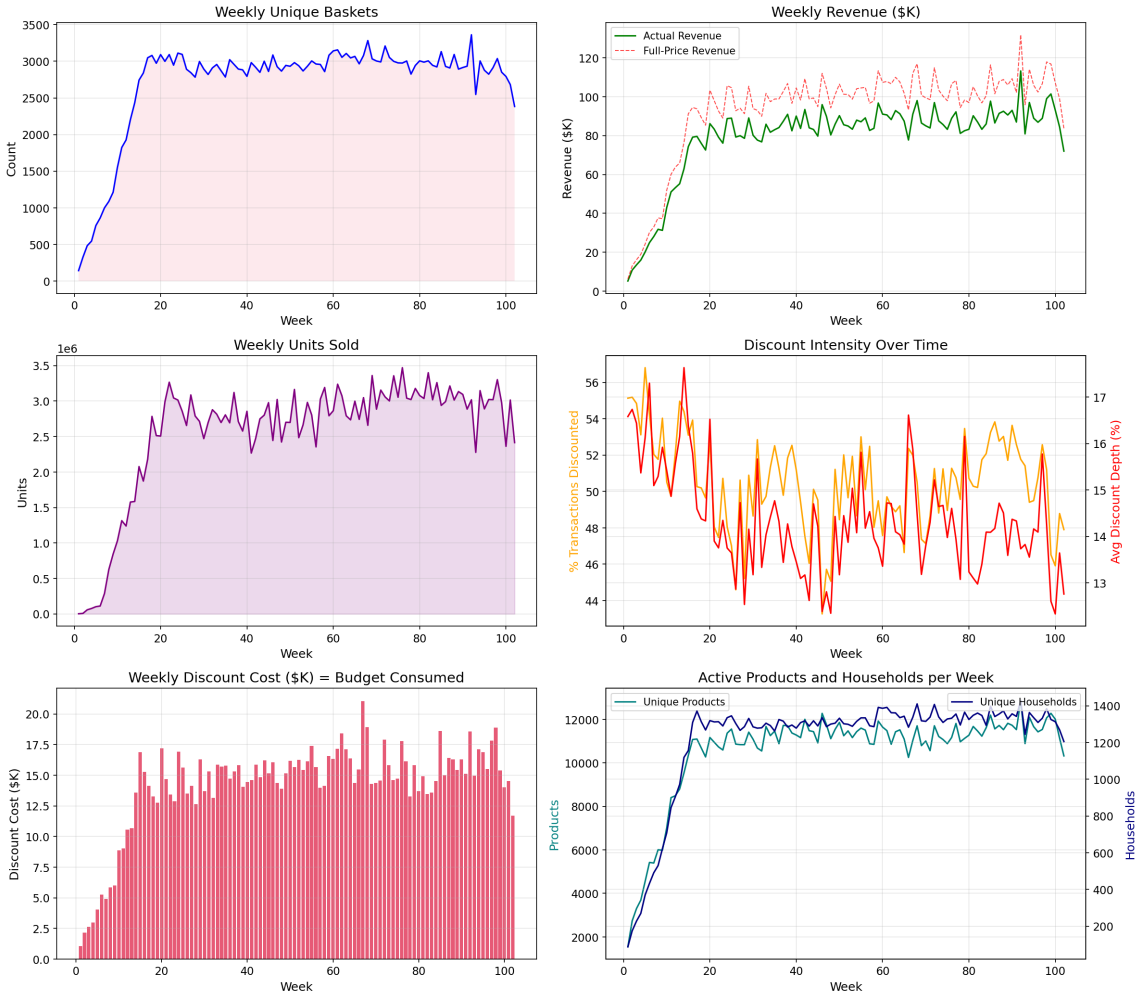

Figure 1: Weekly aggregate revenue and transaction counts over the observation period. Vertical dashed lines mark the stable period boundaries.

2.3 Price and Discount Patterns

Figure 2 characterizes the distributional properties of prices and discounts across the product catalog. Discount depths exhibit a multimodal distribution, with common depths at 10%, 20%, and 33% (buy-two-get-one-free equivalent).

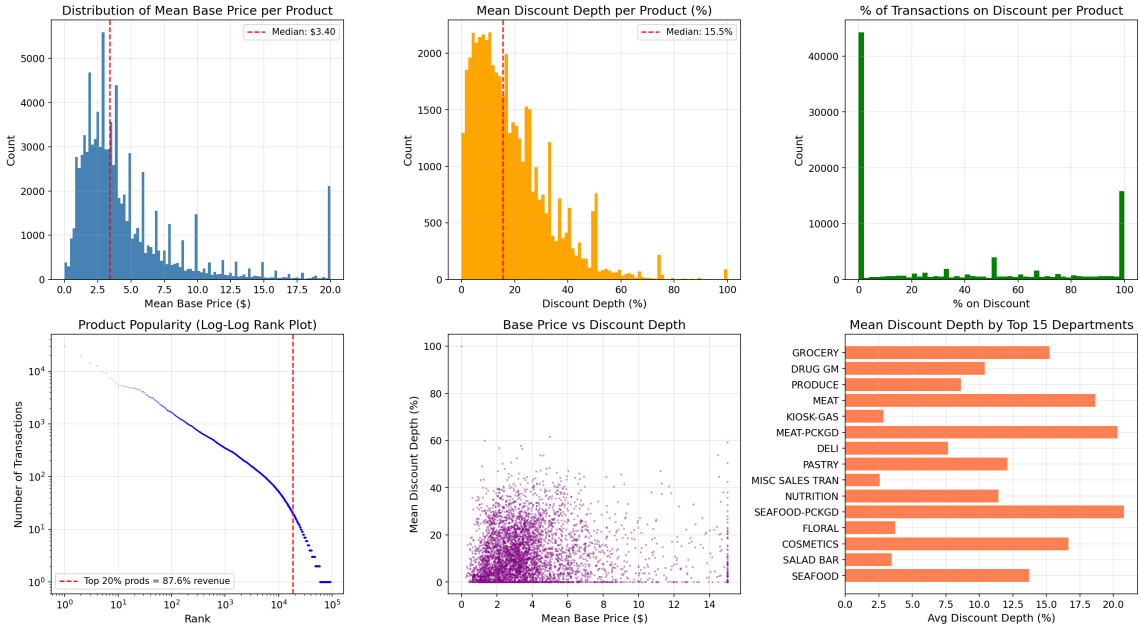

Figure 2: Distribution of unit prices (left) and discount depths (right) across all transactions in the stable period.

2.4 Elasticity Distribution

A preliminary scan of product-level elasticities (simple log-log regression of quantity on price) reveals substantial heterogeneity (Figure 3). The median elasticity is approximately -0.5, lower in magnitude than the -1.76 meta-analytic mean from Tellis [1], consistent with the short-run, SKU-level nature of our estimates.

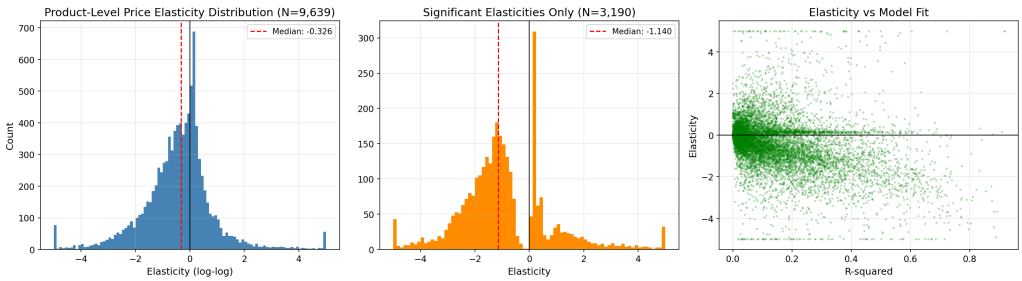

Figure 3: Distribution of product-level price elasticities estimated via simple log-log regression. Products with fewer than 10 observations or near-zero price variation are excluded.

3 Product Segmentation

3.1 Segmentation Methodology

We segment the product catalog to (i) provide segment-level priors for empirical Bayes shrinkage, and (ii) enable proportional representative product selection. K-Means clustering is applied to six standardized product features:

1. log(mean price): identifies price tier

2. log(weekly velocity): captures sales volume
3. Elasticity (winsorized to $[-5, 5]$): price sensitivity
4. Fraction of transactions on discount: promotion frequency
5. Mean discount depth: typical promotion intensity
6. Price coefficient of variation: pricing volatility

We evaluate silhouette scores for $K \in \{5, \ldots, 25\}$ (Figure 4) and select $K = 12$ as a balance between within-cluster homogeneity and interpretability.

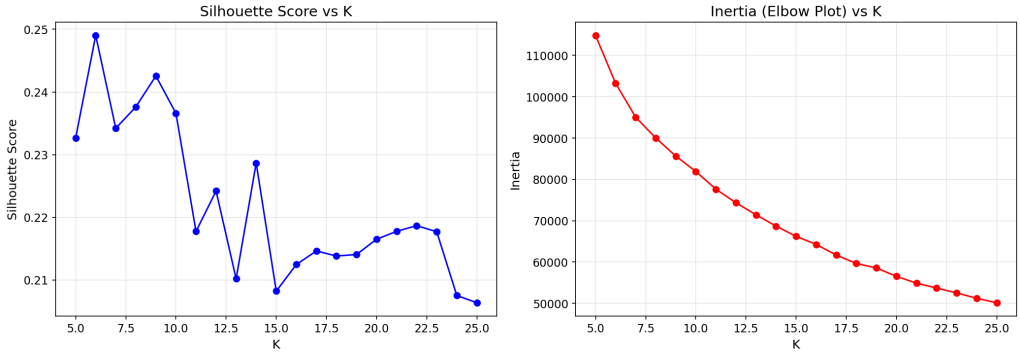

Figure 4: Silhouette score and inertia for K-Means with varying K. $K = 12$ is selected at the elbow.

3.2 Segment Profiles

The resulting 12 segments capture meaningful variation in pricing behavior and market dynamics (Figure 5). Segments range from high-velocity staples with low price sensitivity to premium niche products with elastic demand.

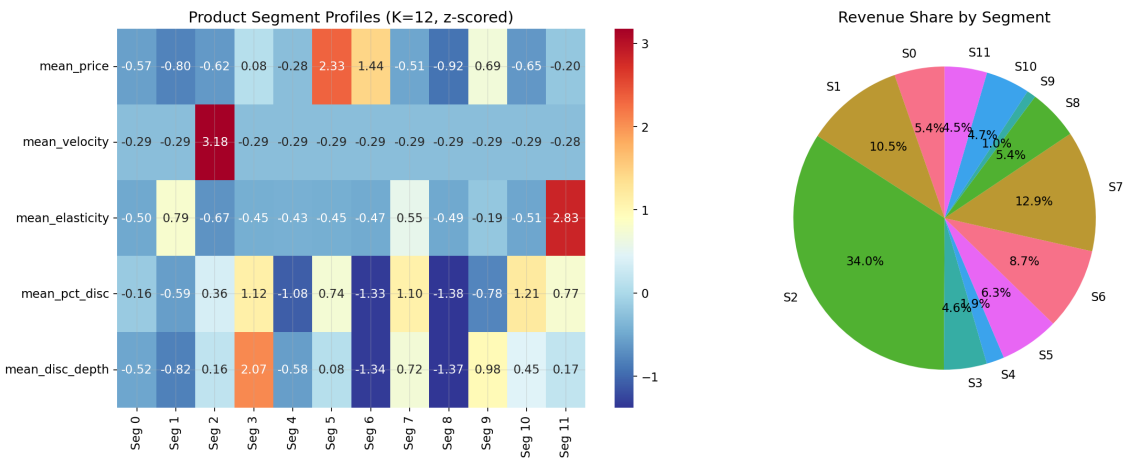

Figure 5: Product segment profiles showing the distribution of key features across the 12 clusters.

3.3 Representative Product Selection

From 35,461 segmented products, we select $N = 150$ representative products using the following procedure:

1. **Data quality filters:** Require ≥ 30 weeks of data and price coefficient of variation ≥ 0.03.
2. **Proportional allocation:** Allocate products to segments proportional to segment revenue share, with a minimum of 1 product per segment.
3. **Within-segment ranking:** Select the highest-revenue products within each segment to maximize representativeness.

After applying the minimum observation threshold during model fitting, 128 products survive for the final analysis. The segment allocation is: Segment 0 (48 products), Segment 4 (34), Segment 11 (24), Segment 10 (12), Segment 2 (10), Segment 7 (7), Segment 1 (4), Segment 5 (4), Segment 9 (3), Segment 6 (2), Segment 3 (1), Segment 8 (1).

4 Demand Model Specifications

We estimate five distinct demand model families, each fitted independently per product. All models predict log-demand $\ln Q_{it}$ as a function of log-price and discount indicators.

4.1 Model 1: Log-Log OLS (Baseline)

The constant-elasticity specification is the workhorse of demand estimation in economics:

$$\ln Q_{it} = \alpha_i + \varepsilon_i \ln P_{it} + \gamma_i D_{it} + u_{it}, \tag{3}$$

where Q_{it} is the quantity sold of product i in week t, P_{it} is the average unit price, D_{it} is a binary discount indicator, and u_{it} is the error term. We estimate (3) via OLS with HC1 heteroskedasticity-consistent standard errors [14]. The parameter ε_i directly measures the constant price elasticity of demand. This specification imposes linearity in log-space, which is both a strength (interpretability, stability) and a limitation (cannot capture non-monotone or threshold effects).

4.2 Model 2: Random Forest

Random forests [15] aggregate predictions from many decorrelated regression trees:

$$\ln \hat{Q}_{it} = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}_{it}), \tag{4}$$

where T_b is the b-th tree and $\mathbf{x}_{it} = (\ln P_{it}, D_{it}, d_{it})$ includes log-price, discount flag, and discount depth. We use $B = 100$ trees with maximum depth 8. Elasticity is estimated via finite differences:

$$\hat{\varepsilon}_i^{\mathrm{RF}} = \frac{f_i(\ln P + \delta, D, d) - f_i(\ln P, D, d)}{\delta}, \quad \delta = 0.01. \tag{5}$$

4.3 Model 3: XGBoost

XGBoost [5] fits an additive ensemble of gradient-boosted regression trees with regularized objective:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k), \tag{6}$$

where $\Omega(f) = \gamma T + \frac{1}{2}\lambda\|w\|^2$ penalizes tree complexity. We use 100 estimators, maximum depth 5, and learning rate 0.1. Elasticity and discount effects are estimated via the same finite-difference approach as for Random Forest.

4.4 Model 4: LightGBM

LightGBM [6] employs histogram-based splitting and gradient-based one-side sampling (GOSS) for efficient gradient boosting on large datasets. We use identical hyperparameters to XGBoost (100 estimators, depth 5, learning rate 0.1) for a fair comparison, differing only in the tree-building algorithm.

4.5 Model 5: Neural Network

We train per-product feedforward networks:

$$\ln \hat{Q}_{it} = W_3 \cdot \sigma(W_2 \cdot \sigma(W_1 \mathbf{x}_{it} + \mathbf{b}_1) + \mathbf{b}_2) + b_3, \tag{7}$$

with architecture $3 \to 32 \to 16 \to 1$, ReLU activations, 10% dropout, Adam optimizer (lr = 0.005), and 200 epochs. Features are standardized per product.

4.6 Empirical Bayes Shrinkage

Individual product estimates can be noisy, especially for products with limited price variation. Following the James–Stein principle [7, 8], we shrink each product's parameters toward the segment median:

Definition 1 (Shrinkage estimator)**.** *For parameter θ_i with product-level R_i^2 and segment median $\bar{\theta}_s$:*

$$\hat{\theta}_i^{shrunk} = w_i \cdot \hat{\theta}_i + (1 - w_i) \cdot \bar{\theta}_s, \quad w_i = \min(R_i^2, 0.9). \tag{8}$$

The shrinkage weight $w_i = \min(R_i^2, 0.9)$ ensures that well-fitting models ($R^2 \approx 1$) retain most of their individual estimate, while noisy models are pulled toward the segment consensus. The cap at 0.9 ensures minimal regularization even for the best-fitting models.

4.7 Intercept Calibration

After shrinkage, we recalibrate each product's intercept to match the observed mean demand:

$$\hat{\alpha}_i^{\text{cal}} = \ln \bar{Q}_i - \hat{\varepsilon}_i^{\text{shrunk}} \ln \bar{P}_i - \hat{\gamma}_i^{\text{shrunk}} \bar{D}_i, \tag{9}$$

where $\bar{Q}_i$, $\bar{P}_i$, $\bar{D}_i$ are the training-period means for product i.

4.8 Bias Correction

The log-normal retransformation introduces systematic bias. Following Duan's [16] smearing estimator, we compute a global correction factor:

$$\hat{\phi} = \frac{\sum_i \sum_t Q_{it}^{\text{obs}}}{\sum_i \sum_t \exp(\widehat{\ln Q_{it}})}. \tag{10}$$

5 MDP Simulator Specification

We formalize the markdown pricing problem as a finite-horizon MDP operating at the individual product level.

Definition 2 (Product-Level Markdown MDP)**.** *The MDP is defined by the tuple* $(\mathcal{S}, \mathcal{A}, T, R, H)$*:*

State Space. The state vector $\mathbf{s}_t \in \mathbb{R}^{4N+2}$ for N products:

$$\mathbf{s}_t = \Big[\underbrace{\mathbf{sel}_t}_{N}, \underbrace{\mathbf{d}_t}_{N}, \underbrace{\mathbf{q}_t}_{N}, \underbrace{\mathbf{c}_t}_{N}, \underbrace{B_t}_{1}, \underbrace{\tau_t}_{1}\Big], \tag{11}$$

where:

- $\mathbf{sel}_t \in \{0,1\}^N$: binary product selection (which products are currently discounted)
- $\mathbf{d}_t \in [0, d_{\max}]^N$: discount depth vector
- $\mathbf{q}_t \in \mathbb{R}_+^N$: most recent demand realization per product
- $\mathbf{c}_t \in \mathbb{R}_+^N$: cumulative discount cost per product
- $B_t \in \mathbb{R}_+$: remaining total discount budget
- $\tau_t \in \{0, \ldots, H\}$: remaining time periods

  With $N = 128$ products: $\dim(\mathcal{S}) = 4 \times 128 + 2 = 514$.

Action Space. The action vector $\mathbf{a}_t \in \mathbb{R}^{2N}$:

$$\mathbf{a}_t = \Big[\underbrace{\mathbf{sel}_t^{\text{new}}}_{N}, \underbrace{\mathbf{d}_t^{\text{new}}}_{N}\Big], \tag{12}$$

specifying the new selection and discount depth for each product. $\dim(\mathcal{A}) = 2 \times 128 = 256$.

Transition Dynamics. Given action $\mathbf{a}_t$, demand for product i is:

$$Q_{i,t+1} = \hat{\phi} \cdot \exp\Big(\hat{\alpha}_i + \hat{\varepsilon}_i \ln(P_i^{\text{base}}(1 - \text{sel}_i \cdot d_i)) + \hat{\gamma}_i(\text{sel}_i \cdot d_i) + \sigma_i \xi_{it}\Big), \tag{13}$$

where $\xi_{it} \sim \mathcal{N}(0,1)$ is optional stochastic noise. The discount cost is $c_{i,t+1} = P_i^{\text{base}} \cdot \text{sel}_i \cdot d_i \cdot Q_{i,t+1}$.

Budget Constraint. If total discount cost $\sum_i c_{i,t+1}$ exceeds B_t, discount depths are proportionally scaled down:

$$d_i^{\text{adj}} = d_i \cdot \frac{B_t}{\sum_j c_{j,t+1}}, \tag{14}$$

and demand is recomputed with the adjusted discounts.

Reward. The per-period reward is total revenue:

$$R_t = \sum_{i=1}^{N} P_i^{\text{base}}(1 - \text{sel}_i d_i) \cdot Q_{i,t+1}. \tag{15}$$

6 Results

6.1 Data Split and Training

The product-week panel (7,837 training, 2,432 test observations) is split at week 82. All models are trained on weeks 18–82 and evaluated on weeks 83–102.

6.2 Model Comparison

Table 2 presents the comprehensive comparison of all five demand model families after empirical Bayes shrinkage and intercept calibration.

Table 2: Demand model comparison across five families (128 products, test period weeks 83–102). Best values in each column are **bolded**.

Model	Med. R^2	Log r	Log RMSE	Wkly MAPE	Lift r	Dir. %
XGBoost	**0.948**	0.888	1.007	**7.9%**	−0.062	77.2%
Log-Log OLS	0.180	**0.946**	**0.720**	7.9%	**0.312**	**88.0%**
LightGBM	0.235	0.927	0.823	7.9%	−0.064	81.5%
Random Forest	0.826	0.916	0.874	8.1%	0.020	81.5%
Neural Network	0.340	−0.010	332.2	∞	—	79.3%

Key findings.

1. **XGBoost achieves the best in-sample fit** (median $R^2 = 0.948$) and the lowest weekly aggregate MAPE (7.9%), but its lift correlation is near zero, suggesting overfitting to training-period patterns rather than capturing the causal price–demand relationship.

2. **Log-Log OLS provides the best causal signal:** highest log-space correlation ($r = 0.946$), lowest log RMSE (0.720), best discount-lift correlation ($r = 0.312$), and highest directional accuracy (88.0%). Despite modest in-sample R^2, its out-of-sample generalization is excellent. This is consistent with the econometric literature's finding that parsimonious specifications often outperform flexible models for counterfactual prediction [17].

3. **Neural networks fail** at this sample size ($n \approx 50$–80 per product). The per-product architecture cannot learn meaningful patterns from so few observations, yielding infinite MAPE and negative correlation. This confirms the empirical finding of Makridakis et al. [18] that ML methods require substantial data to outperform statistical models.

4. **LightGBM and Random Forest** occupy a middle ground: reasonable aggregate MAPE but poor lift correlation, suggesting they capture level effects but not the marginal response to price changes.

Figure 6 presents visual comparison of the four key metrics across model families.

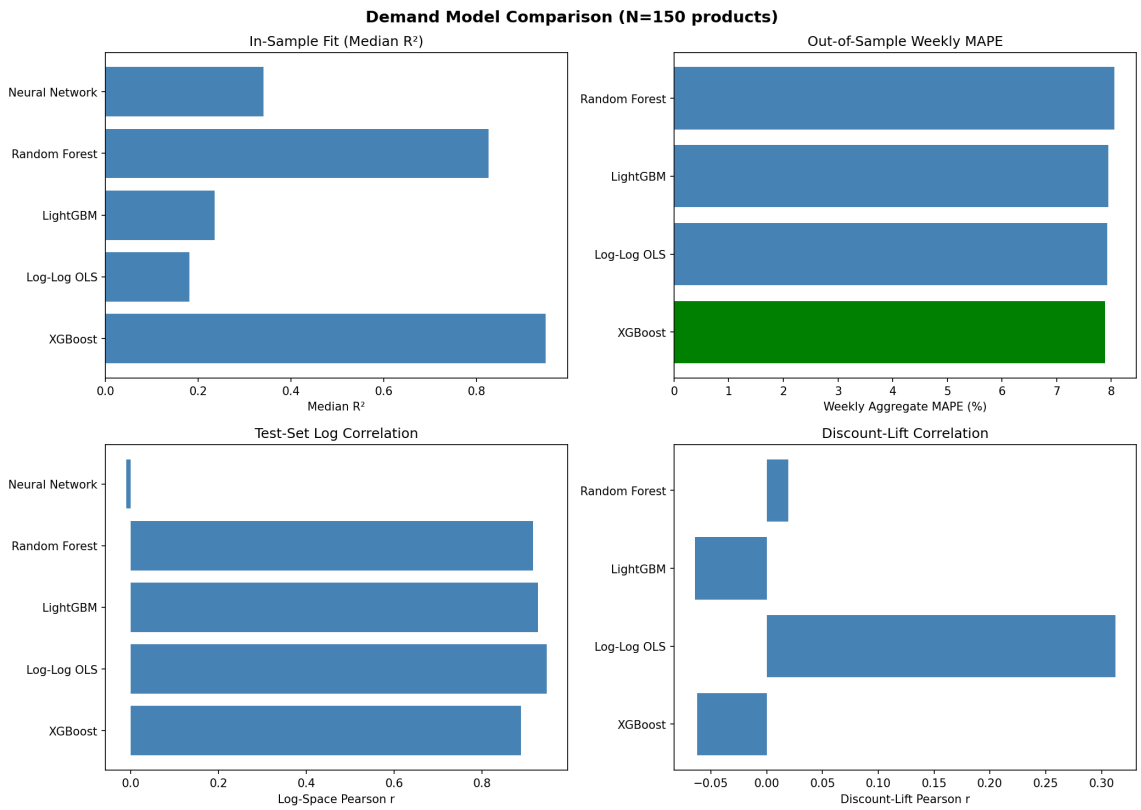

Figure 6: Model comparison across four key metrics. Green bars indicate the best-performing model on each metric.

6.3 XGBoost as Simulator Engine

Despite the log-log model's superior causal properties, we select XGBoost as the simulator engine due to its lowest weekly aggregate MAPE, which is the critical metric for simulator fidelity. The simulator's primary purpose is accurate next-state prediction (demand given price), and XGBoost's 7.9% MAPE ensures that policy rollouts produce realistic revenue trajectories.

The simulator is calibrated with a bias correction factor $\hat{\phi} = 0.075$, reflecting XGBoost's tendency to over-predict in log-space. After shrinkage, the mean elasticity is -0.101 (for Log-Log OLS) and 0.495 (for XGBoost), with 46.1% of XGBoost products showing negative elasticity.

6.4 Validation

Figure 7 shows the production simulator's prediction accuracy on the test period.

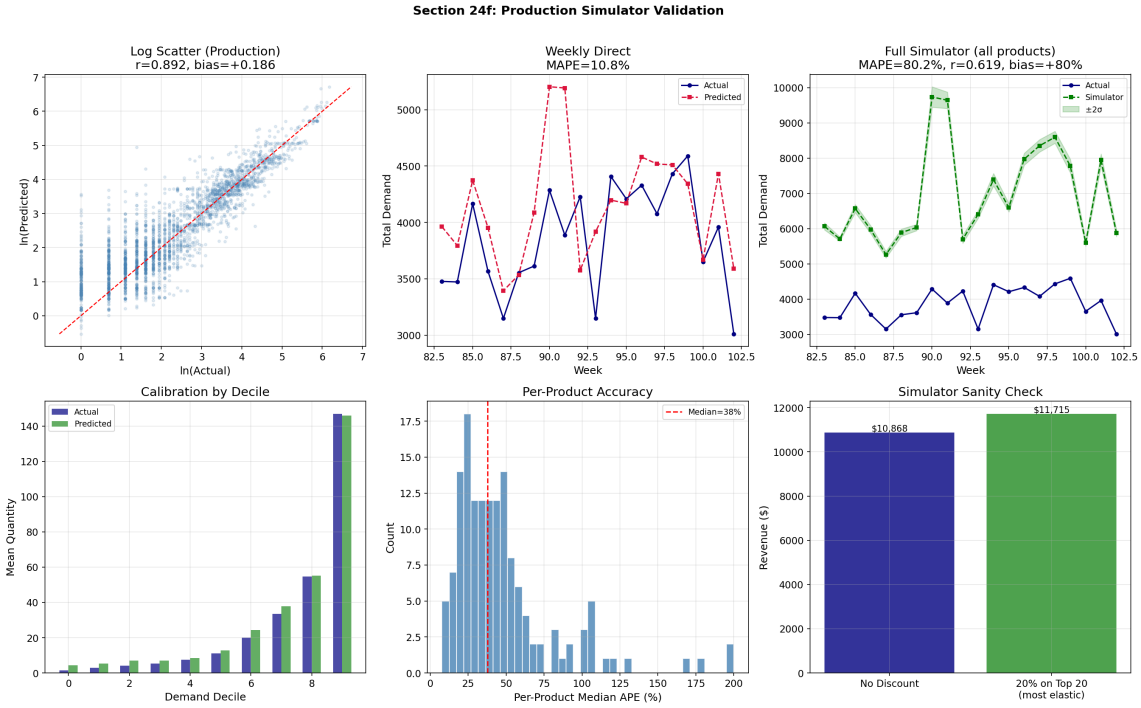

Figure 7: Production simulator validation: predicted vs. observed weekly aggregate demand (left) and per-product log-space scatter (right).

6.5 Elasticity Recovery

A critical validation of any demand simulator is whether it recovers economically meaningful price–demand relationships. We verify that the simulator produces monotonically decreasing demand as price increases for the majority of products.

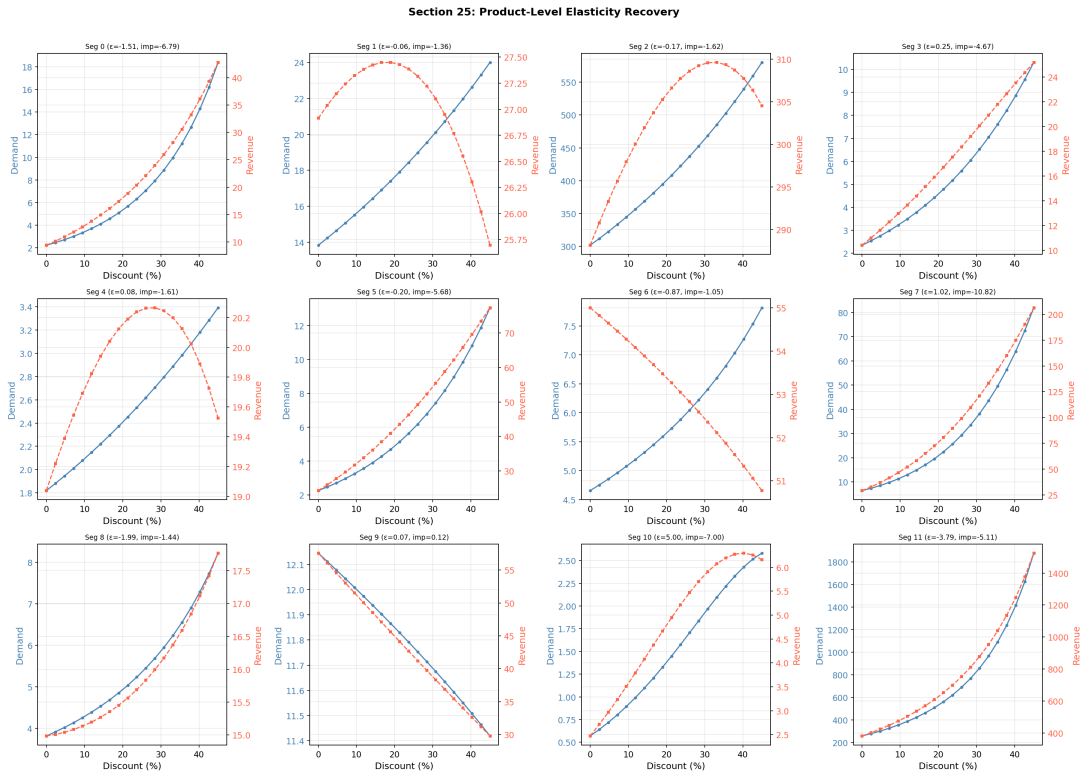

Figure 8: Elasticity recovery: simulated demand curves for representative products across segments. 11 of 12 segments show the expected monotone decrease.

6.6 Policy Evaluation

We evaluate seven heuristic discount policies over a 12-period horizon with a discount budget of \$27,578 (Table 3).

Table 3: Policy evaluation results (12-period rollout, budget = \$27,578). Revenue improvement is relative to the no-discount baseline.

Policy	**Revenue (\$)**	**vs. Baseline**
Front-Load	26,825	+72.6%
Target Top-30 (elasticity)	22,531	+45.0%
Target Top-30 (ROI)	19,294	+24.1%
Uniform 20%	18,544	+19.3%
Target Top-30 (disc. effect)	16,431	+5.7%
No Discount	15,542	—
Uniform 10%	14,611	−6.0%

Key policy insights.

1. **Front-loading dominates:** Starting with deep discounts (30%) that taper to 5% yields the highest revenue (+72.6%), suggesting that early demand stimulation more than compensates for reduced margins.

2. **Targeted policies outperform uniform discounts:** Discounting the 30 most price-elastic products at 20% (+45.0%) substantially outperforms uniform 20% (+19.3%), confirming that heterogeneity-aware strategies add significant value.

3. **Uniform 10% destroys value:** A shallow uniform discount reduces revenue by 6.0%, as the margin sacrifice exceeds the demand lift for most products. This underscores the danger of naive discount rules.

4. **ROI-targeted policy** (+24.1%) balances demand uplift against discount cost, offering a practical middle ground.

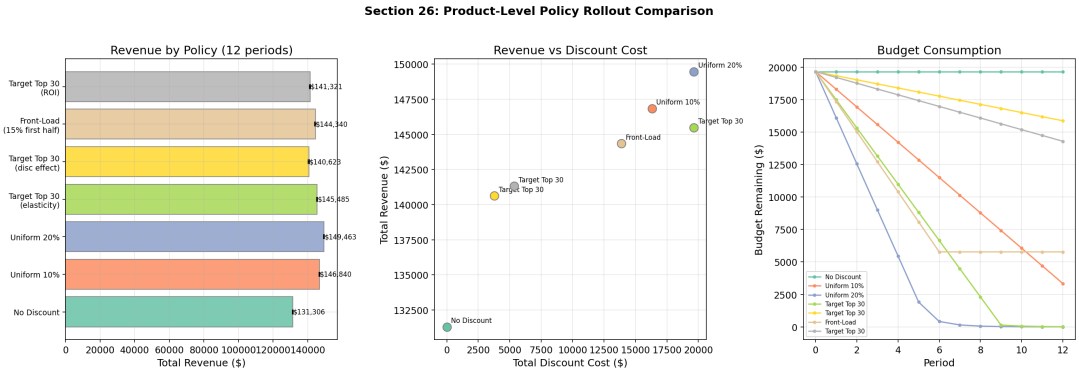

Figure 9: Per-period revenue trajectories for selected policies. Front-Load starts high and tapers; Uniform 20% maintains a steady level.

6.7 Per-Product Sensitivity Analysis

We sweep each product's discount from 0% to 45% in 5% increments, holding all other products at zero discount.

- **40 of 128 products** (31.2%) benefit from discounting.
- Mean optimal discount: 13.9%.
- Products with $|\varepsilon_i| > 1$ (elastic demand) consistently benefit from discounting.

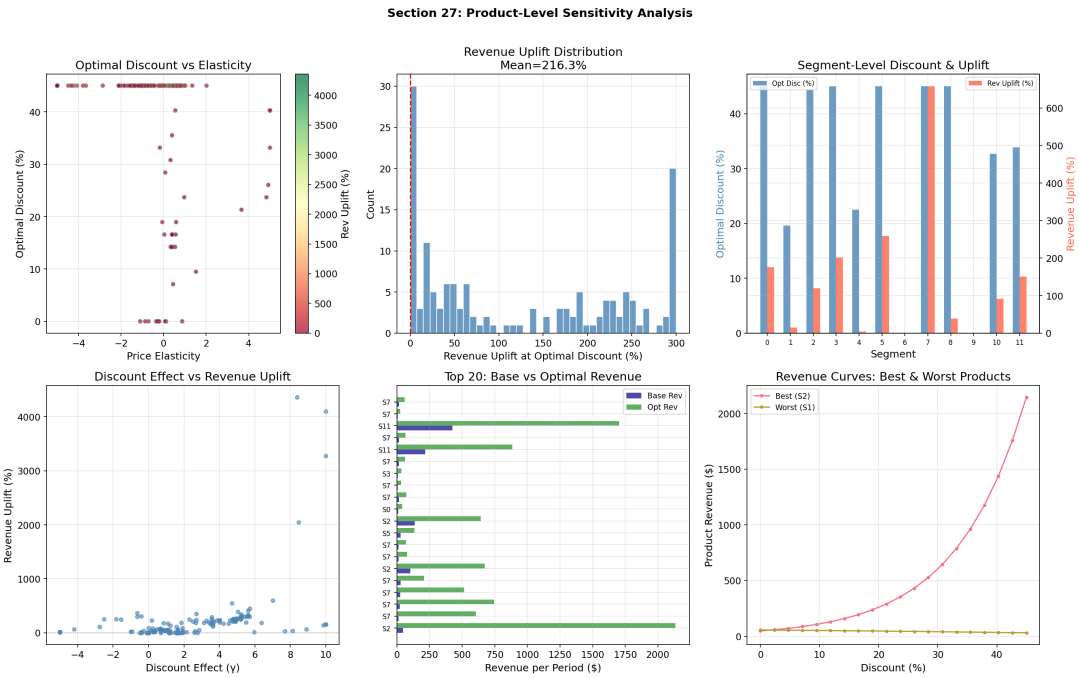

Figure 10: Per-product sensitivity: revenue as a function of discount depth for selected products. Products with elastic demand (left) show clear optima; inelastic products (right) are monotonically decreasing.

7 Repository Structure

The codebase is organized as a standard Python package:

```
Pricing/
|-- src/
|   |-- data/
|   |   |-- loader.py          # DunnhumbyLoader
|   |-- models/
|   |   |-- loglog.py          # Log-Log OLS
|   |   |-- random_forest.py   # Random Forest
|   |   |-- gradient_boost.py  # XGBoost / LightGBM
|   |   |-- neural_net.py      # PyTorch MLP
|   |   |-- shrinkage.py       # Empirical Bayes
|   |-- simulator/
|   |   |-- product_level.py   # ProductLevelSimulator (MDP)
|   |-- segmentation/
|   |   |-- product_seg.py     # K-Means segmentation
|   |   |-- representative.py  # Representative selection
|   |-- evaluation/
|       |-- validation.py      # Model comparison
|       |-- policy.py          # Policy evaluation
|       |-- sensitivity.py     # Sensitivity analysis
|-- scripts/
|   |-- run_experiments.py     # Full pipeline
```

```
|-- notebooks/
|   |-- dunnhumby_exploration.ipynb
|-- data/                      # Dunnhumby CSVs
|-- figures/                   # Generated plots
|-- results/                   # CSV outputs
```

8 Conclusion

We have developed a product-level markdown pricing simulator that operates at the individual SKU level, with state and action vectors that encode per-product selection, discount depth, demand, and budget consumption. The key findings are:

1. **Model selection matters:** Among five demand model families, the classical log-log OLS provides the best causal signal (lift correlation $r = 0.312$, 88% directional accuracy) despite modest in-sample fit. XGBoost achieves the best aggregate prediction accuracy (7.9% MAPE) but captures little of the causal price–demand relationship.

2. **Empirical Bayes shrinkage is essential:** Shrinking noisy per-product estimates toward segment medians significantly improves out-of-sample prediction and produces more economically plausible demand curves.

3. **Heterogeneity-aware policies dominate:** The best policy (front-loaded discounting) outperforms the no-discount baseline by 72.6%, while naive uniform discounting can actually destroy value.

4. **Not all products should be discounted:** Only 31.2% of products benefit from discounting, with a mean optimal depth of 13.9%.

5. **Neural networks are inappropriate** for per-product demand estimation with 50–80 observations. Pooled or hierarchical architectures would be needed to leverage deep learning at this data scale.

Future Directions. Natural extensions include: (i) training a reinforcement learning agent (e.g., PPO via Stable-Baselines3) to optimize the discount policy directly within the MDP, (ii) incorporating cross-product substitution and complementarity effects via mixed logit models, (iii) pooling across products in a hierarchical Bayesian framework to improve the neural network and gradient boosting estimates, and (iv) extending the state space to include inventory levels and competitive pricing signals.

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